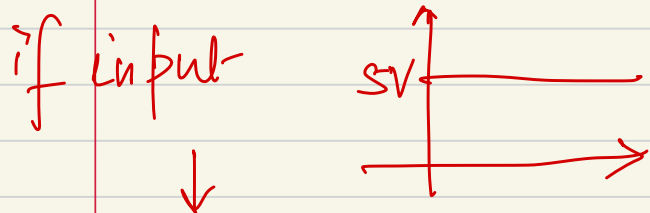


Tutorial 1

1) PWM - 100 Hz, $T = 0.01 \text{ s}$ $\rightarrow 10 \text{ ms}$
0,5 V

A) $P_{\text{reg, av}} = 0.5 \text{ W}$, Duty cycle of PWM

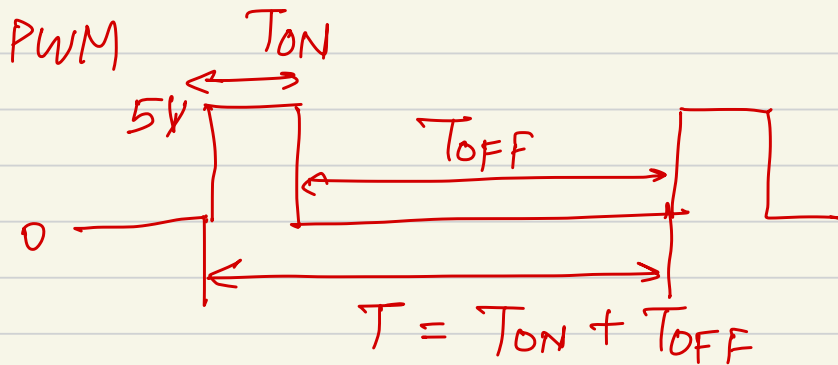


$P = \frac{5^2}{10} = 2.5 \text{ W}$

$P_{\text{reg}} = 0.5 \text{ W} = D \times 2.5 \text{ W}$

$D = \frac{1}{5} = 0.2 = 20\%$

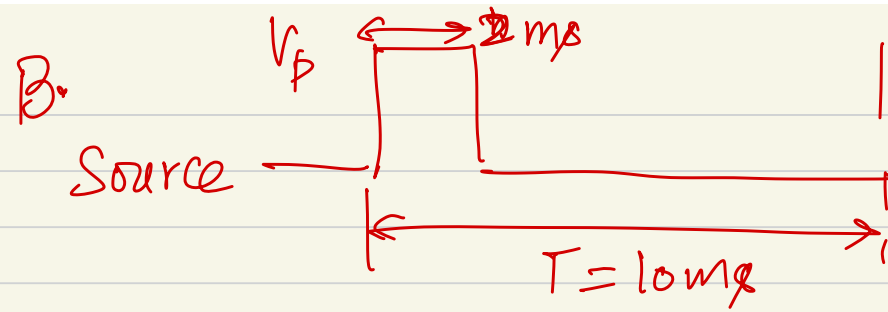
$T_{\text{ON}} = DT = 0.2 \times 10 \text{ ms} = 2 \text{ ms}$



Duty cycle $= \frac{T_{\text{ON}}}{T} = D$

$P_{\text{avg.}} = \frac{1}{T} \int_0^T \frac{U^2(t)}{R} dt$

$= \frac{1}{T} \int_0^T \frac{5^2}{10} dt = D \times \frac{5^2}{10} = D \times 2.5$



DC Voltmeter $\rightarrow$ average value of the signal

$$V_{\text{avg}} = \frac{1}{T} \int_0^T v(t) dt$$

$$= \frac{1}{T} \int_0^{DT} V_p dt = DV_p = 0.2 \times 5 = 1\text{ V}$$

P_{avg} using V_{avg}

$$= \frac{1^2}{10} = 0.1\text{ W}$$

C.

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^{DT} v^2(t) dt} = \sqrt{\frac{1}{T} \times DT \times 5^2} = 5\sqrt{D} = 5 \times \sqrt{0.2} = 2.236\text{ V}$$

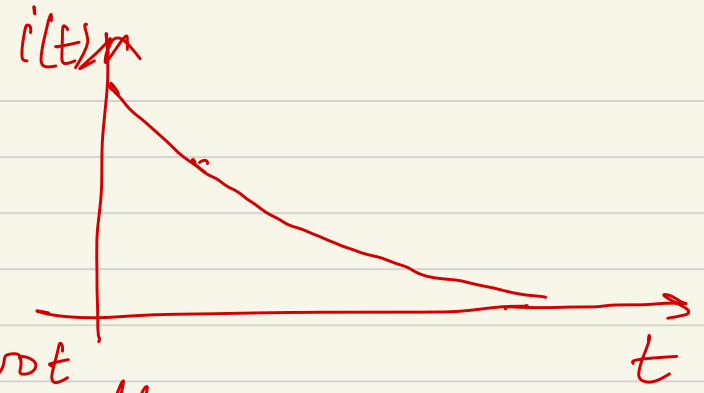
P_{avg} using V_{rms}

$$= \frac{(5\sqrt{2})^2}{10} = \frac{25 \times 2}{10} = 0.5\text{ W}$$

2

$$i(t) = 20\text{K} e^{-1000t} \text{ A}, t \geq 0$$

$$R = 10 \Omega$$



$$E = \int_0^{\infty} i^2(t) R dt = 10 \times (20\text{K})^2 \int_0^{\infty} e^{-2000t} dt$$

$$= 10 \times 10 \times 400 \left[\frac{e^{-2000t}}{-2000} \right]_0^{\infty} \text{ J}$$

$$= 4 \times 10^9 \left[0 + \frac{1}{1000} \right] = 4 \times 10^6 \text{ J} = 4 \text{ MJ}$$

$$i(t) = \frac{dQ}{dt}$$

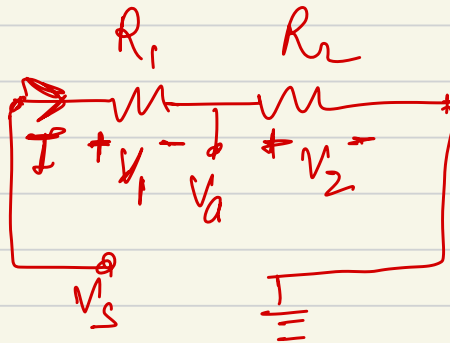
$$Q = \int_0^{\infty} i(t) dt = 20\text{K} \int_0^{\infty} e^{-1000t} dt = \frac{20\text{K}}{1000} = 20 \text{ C}$$

Q.3

$$V_s = 10V$$

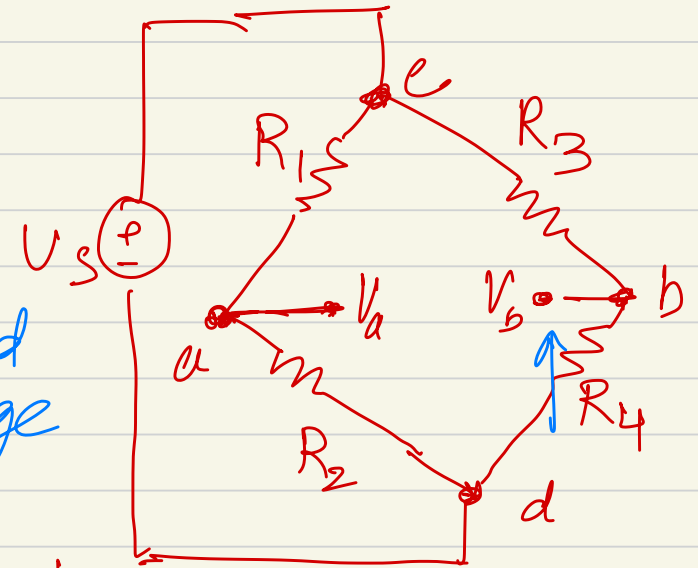
$$R_1 = 1K, R_2 = 2K, R_3 = 5K$$

$$R_x \text{ so that } V_{ab} = V_a - V_b = 0 \\ \Rightarrow V_a = V_b$$



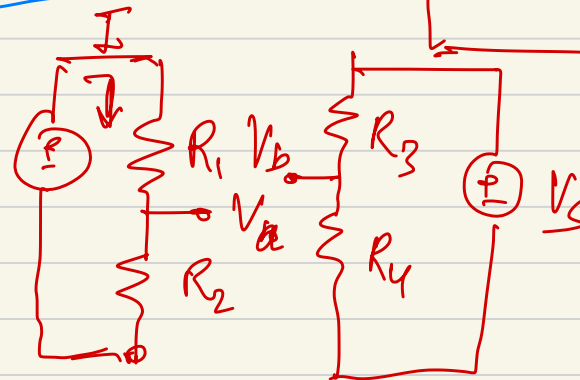
$$\frac{R_2}{R_1 + R_2} = \frac{R_4}{R_3 + R_4} \\ \frac{R_2}{R_1 + R_2} = \frac{R_4}{R_3 + R_4} \\ \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Balanced Bridge



$$R_4 = \frac{R_1 R_3}{R_2}$$

$$V_1 = IR_1, V_2 = IR_2$$



KVL

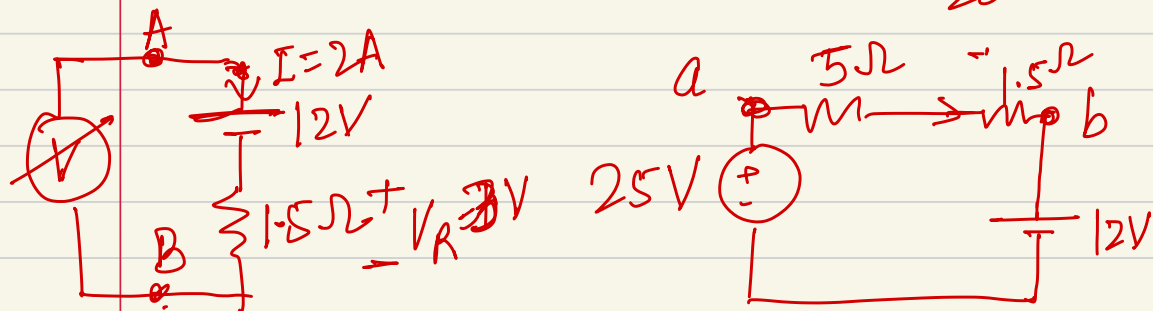
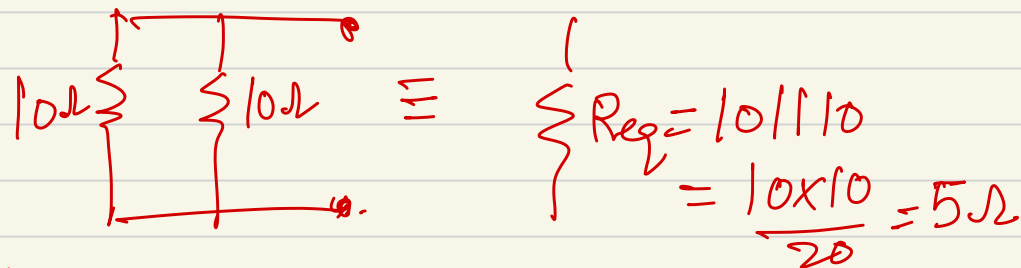
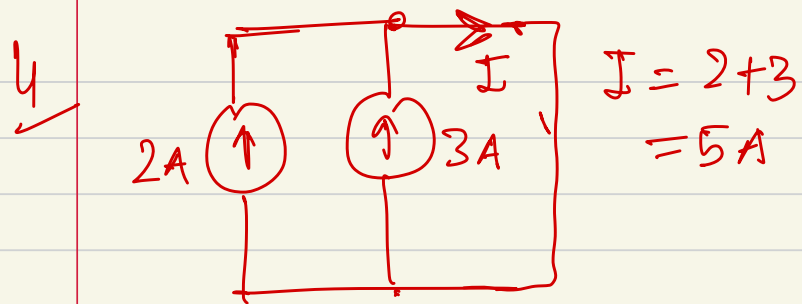
$$V_s = V_1 + V_2, V_s = I(R_1 + R_2)$$

$$I = \frac{V_s}{R_1 + R_2}$$

$$V_2 = V_a = \frac{R_2}{R_1 + R_2} V_s$$

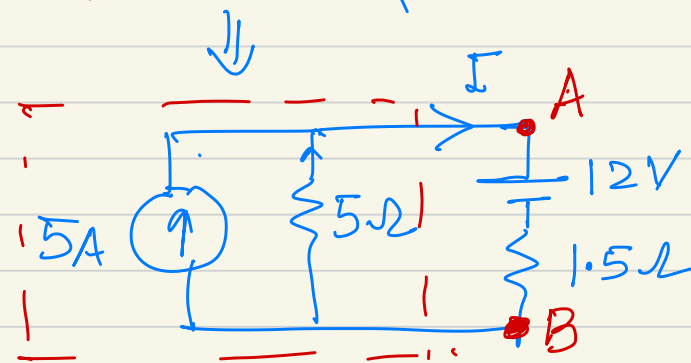
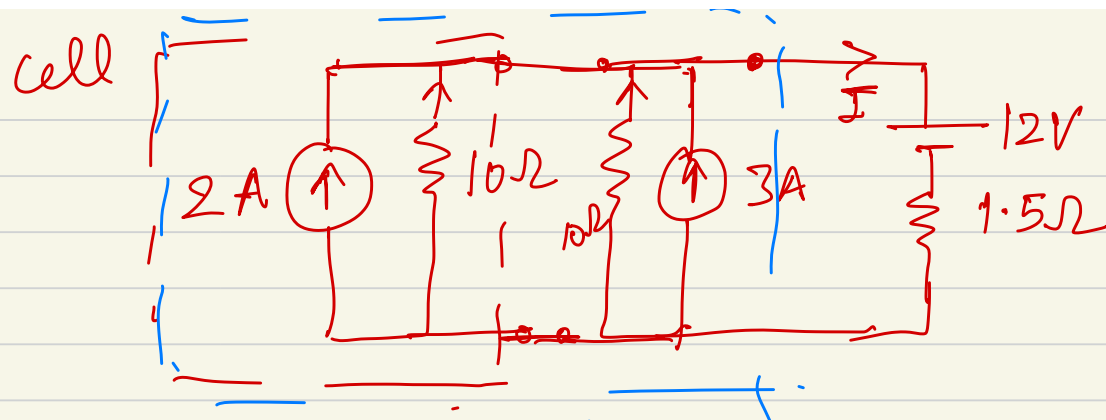
$$V_a = \frac{R_2}{R_1 + R_2} V_s$$

$$V_b = \frac{R_4}{R_3 + R_4} V_s$$



(B) $V_A - 12V - 1.5 \times 2 - V_B = 0$

$V_A - V_B = 12 + 3 = 15V$



(A)

$\Rightarrow I = \frac{25 - 12}{6.5} = \frac{13}{6.5} = 2A$

(C) $Q = 20Ah$

$t = \frac{Q}{I} = \frac{20Ah}{2A} = 10h$

5 KCL nodes: a, b, c, d.

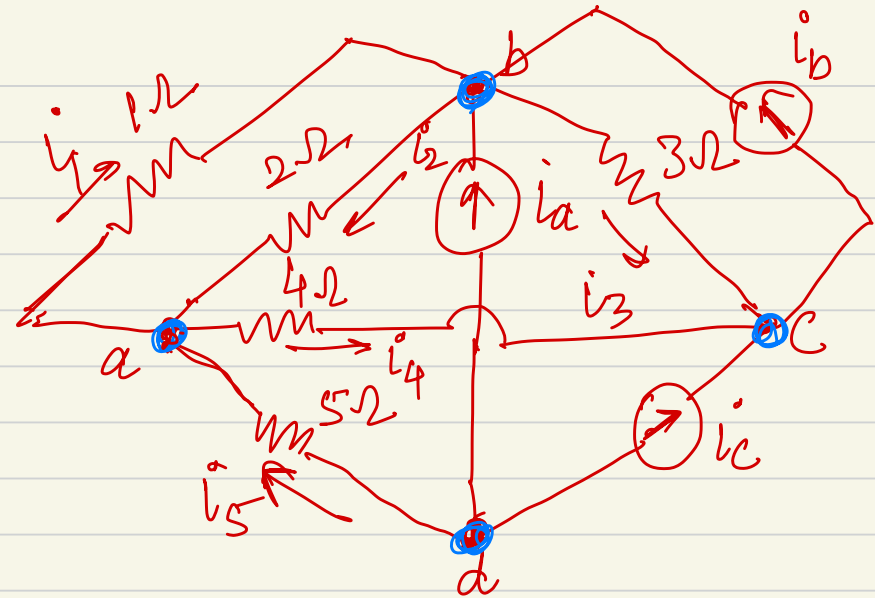
node a leaving $\rightarrow +$
entering $\rightarrow -$

$$i_1 + i_4 - i_2 - i_5 = 0$$

node b $i_2 + i_a - i_1 - i_b = 0$

node c $i_b - i_4 - i_3 - i_c = 0$

node d $i_5 + i_c + i_a = 0$



KVL in loop a, b, c, d

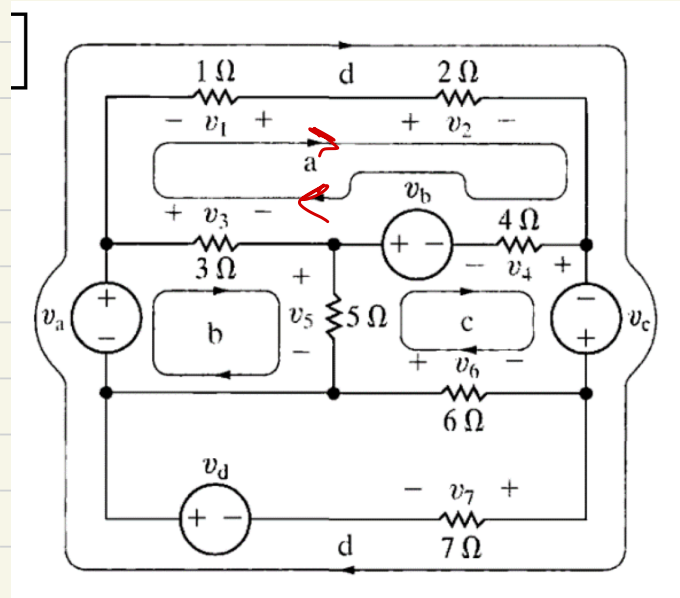
→ mesh is a special loop which does not contain any other loops inside it.

Loop a :-
$$v_1 - v_2 - v_4 + v_b + v_3 = 0$$
$$-v_1 + v_2 + v_4 - v_b - v_3 = 0$$

Loop b :-
$$-v_3 - v_5 + v_a = 0$$

Loop c :-
$$-v_b + v_4 + v_c + v_6 + v_5 = 0$$

Loop d :-
$$v_1 - v_2 + v_c - v_7 + v_d + v_a = 0$$



Q6 $V_s ? = V_{FD} = V_{FE} + V_{ED}$

$V_{FE} = 1.2 \times 5A = 6V = \underline{60V}$

$V_{ED} = 54V$

KCL at node C

$i_{CE} + i_{CB} + i_{CD} + i_{CA} = 0$

$i_{CE} + 1A + 0.5A + 3A = 0$

$i_{CE} = -4.5A$

$i_{EC} = 4.5A$

$V_{EC} = 4.5 \times 4\Omega = 18V$

$V_{ED} = V_{EC} + V_{CD}$
 $= 18 + 36 = 54V$

$i_{ED} = \frac{54}{108} = 0.5A$

$V_{BD} = 12V \quad V_{22} = 6V$

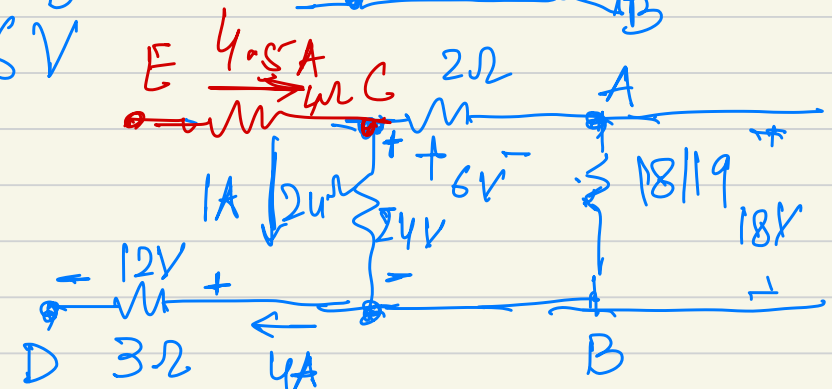
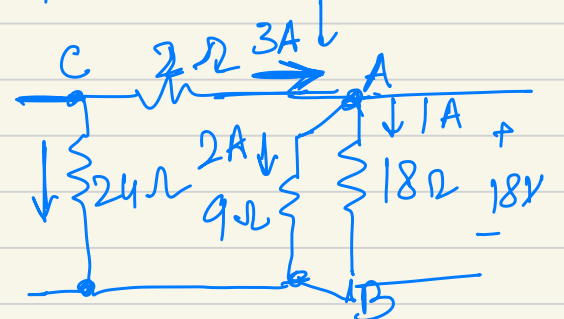
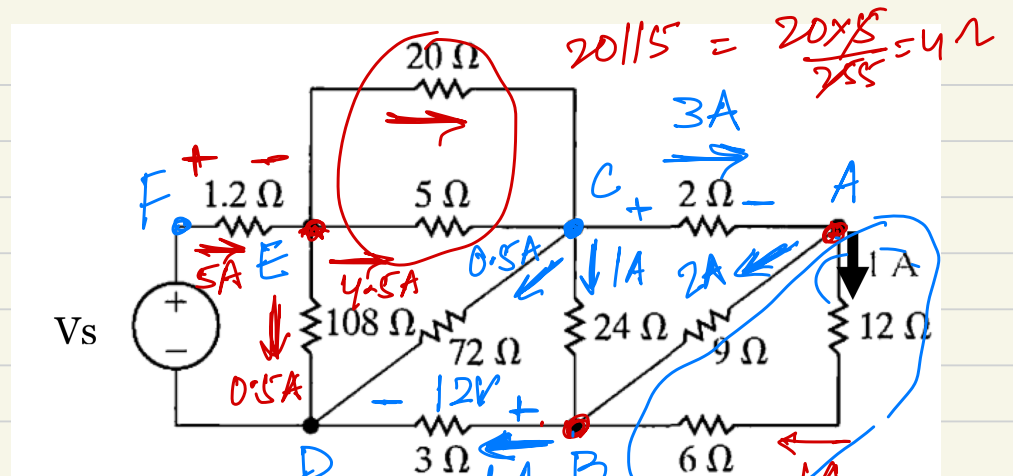
$V_{AB} = 18V = V_A - V_B$

$V_{CA} = V_C - V_A = 6V$

$V_{CB} = V_C - V_B$
 $= 18 + 16 = 24V$

$V_{CD} = V_C - V_D$

$= V_{CB} + V_{BD} = 24 + 12 = 36V, \quad i_{CD} = \frac{36}{72} = 0.5A$

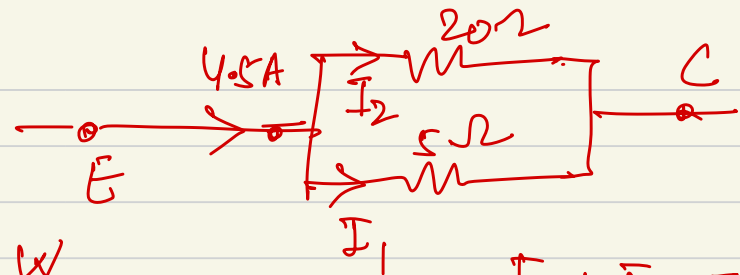


B)

$$P = I_2^2 \times 20 \Omega$$

$$= (0.9)^2 \times 20 \Omega = 0.81 \times 20$$

$$= \underline{8.1 \times 2 = 16.2 \text{ W}}$$



$$I_1 + I_2 = 4.5 \text{ A}$$

$$V_{EC} = 18 \text{ V}$$

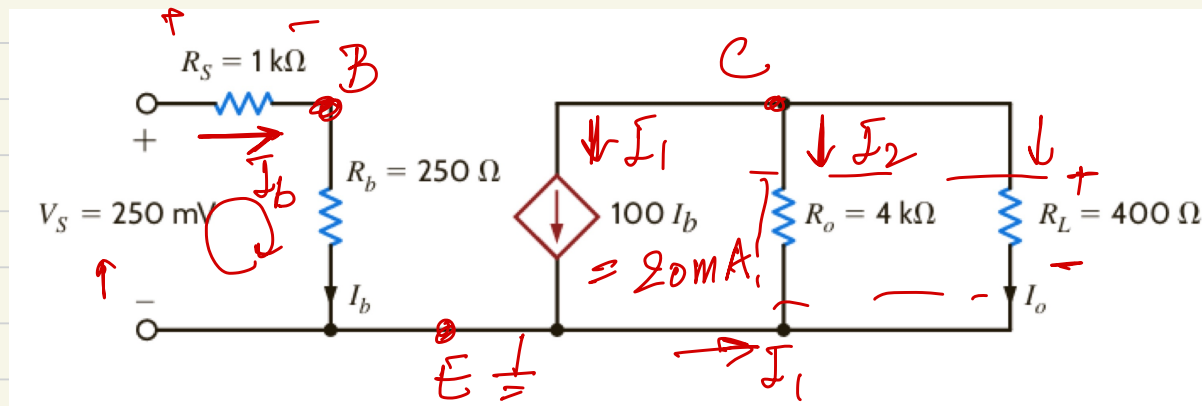
$$I_1 = \frac{18}{5} \text{ A}, I_2 = \frac{18}{20} \text{ A} = 0.9 \text{ A}$$

Q7.

Transistor $\Rightarrow$ CCCS

I_o

$$= -\frac{200}{11} \text{ mA}$$



Current/voltage Dependent current source (CCCS)
[VCCS]

$$250 \text{ mV} - I_b \cdot 1 \text{ k} - 250 \cdot I_b = 0$$

$$\Rightarrow I_b = \frac{250 \text{ mV}}{5 \cdot 1250 \Omega} = 0.2 \text{ mA}$$

KCL at C

$$I_1 + I_2 + I_o = 0$$

$$I_1 = 100 I_b = 20 \text{ mA}$$

$$V_{EC} = \frac{4000}{11} \times 20 \text{ mA} = \frac{80}{11} \text{ V} \Rightarrow V_{CE} = -\frac{80}{11} \text{ V}$$

$$V_{CE} = I_o R_L \Rightarrow I_o = \frac{-\frac{80}{11}}{400} = -\frac{2}{110} \text{ A} = -\frac{2000}{110} \text{ mA}$$

